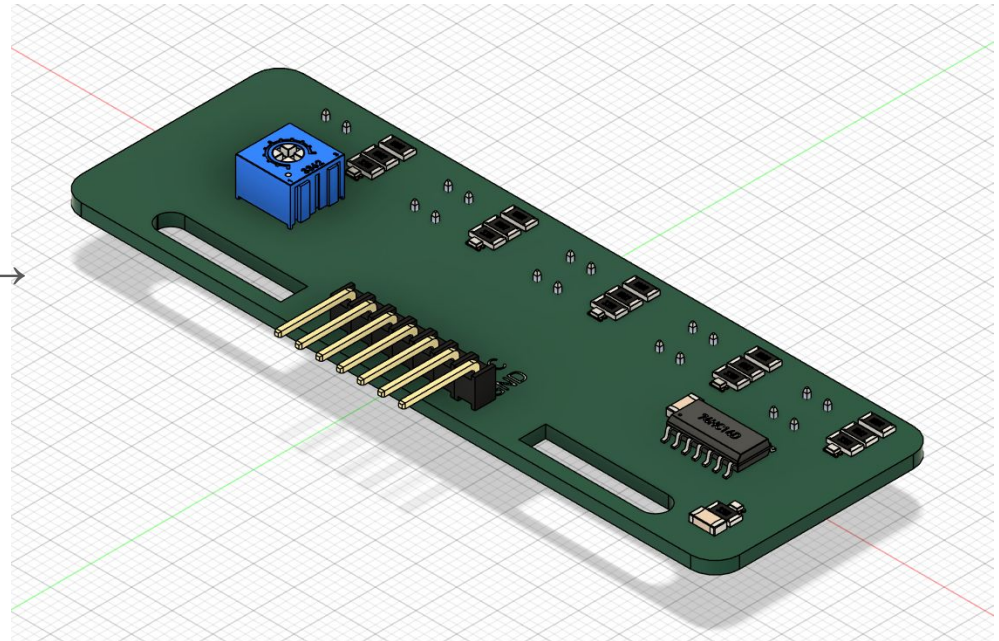


Unit 3: Measurement Workshop

BEFORE THE BELL RINGS

1. Phones put in pouch
2. Backpacks → under chair
3. Open Fusion 360 → File →
Open from Hub → Common
Components → Electronics →
“5 Channel IR Tracker”



TOOL #1: DIGITAL CALIPER

HOLD: thumb on slider, index on frame—like a game controller

ZERO: close jaws → press ZERO → open

UNITS: press mm/inch

MEASURE: flat jaws OUTSIDE, thin tips INSIDE

RULE: measure 3×, write the middle number

TOOL #2: TAPE MEASURE

The foot is LOOSE on purpose!

Inside a box → hook pulls

Outside an edge → hook pushes

The wobble = 1 mm (thickness of the hook)

NEVER trust the first 1 in of the tape!

ACCURACY vs PRECISION

ACCURACY = close to the TRUTH

PRECISION = same number every time

 Accurate + Precise = 3 bullseyes

 Precise only = 3 shots in the wrong hole (How?)

 Accurate only = may not meet tolerances

We want BOTH → 3 measurements + average

Sticky Note Challenge: Measure and record three points on the IR sensor

